

Shubh Mall

Software Engineer

mall.shubhkumar@gmail.com | (604) 618-7535 | Burnaby, BC | [LinkedIn](#)

WORK EXPERIENCE

Schneider Electric Prosumer

Software Engineer / Data Engineer

Aug 2023 – Present

Richmond, BC

- Delivered fleet-wide data insights for 200+ solar inverters through a Databricks medallion pipeline, surfacing real-time performance dashboards in Power BI for cross-functional teams including operations, engineering, and executives; automated data workflows with CI/CD integration and passed internal cybersecurity review for production deployment
- Designed Databricks medallion architecture, processing data from a fleet of 200+ solar inverters into Power BI dashboards; implemented CI/CD pipeline (Github Actions) with dev/prod separation and passed internal cybersecurity review for production release
- Built Python-based manufacturing automation for solar inverters; automated firmware flashing, factory configuration, cloud provisioning, and burn-in testing across 11 stages, reducing manual input by 96% and increasing production line throughput by 50%
- Engineered System Diagnostics Application (Python/Tkinter) for 20 staff; executed remote shell commands via Paramiko SSH routed through an Azure-hosted WireGuard VPN to reach IoT devices behind NAT, cutting diagnostics time by 50%
- Proactively led critical Wi-Fi optimization effort that reduced fleet-wide No Data sites from 6.1% to 0.7% (137 sites); owned end-to-end from root-cause analysis through edge-case testing to accelerated deployment outside standard release cycles
- Led Software Control Tower initiative, consolidating feedback from lab testers, field trial participants, systems teams, and application engineers into executive-level reports on critical fleet issues, bug status, and improvement timelines

Xantrex Technology Inc.

Embedded Firmware Engineer

May 2022 – Aug 2023

Burnaby, BC

- Engineered dual-band WiFi connectivity (2.4/5GHz) with intelligent band selection and country code configuration ensuring reliable connectivity
- Developed udev rules for kiosk touchscreen integration; automated cursor hiding, input detection, and Chromium lifecycle management to conserve power
- Managed Jenkins CI pipeline producing Raspberry Pi production builds across multiple repositories through candidate releases
- Created and executed comprehensive test plans including regression testing; identified and resolved bugs through root cause analysis
- Configured and tested embedded systems in electronics lab environment with high-voltage AC/DC power sources; established test stations for hardware validation

EDUCATION

Simon Fraser University

Bachelor of Applied Science, Computer Science

3.1 GPA | Dota 2 Esports Team Captain | Soccer Team Captain

Apr 2024

Burnaby, BC

TECHNICAL SKILLS

Languages: Python, SQL, Bash, TypeScript, C/C++

Platforms: Linux, Databricks, Azure, Power BI

Frameworks & Libraries: Tkinter, PySpark, React, Django, Pandas, Paramiko, REST APIs

Tools: Docker, Jenkins, Git, GitHub Actions, OpenSSL

Protocols: OCPP, WireGuard, SSH/SFTP, OAuth, Modbus, CAN, WebDAV